

1. Launch one EC2 instance using Amazon Linux 2

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Instances' page displays a list of EC2 instances. The instance 'devopschallenge' (ID: i-0bb8dbdfef417e6a) is in the 'Running' state. Below the list, the details for this instance are shown, including its public IPv4 address (50.19.60.48) and public DNS (ec2-50-19-60-48.compute-1.amazonaws.com).

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
server 2	i-09d8c327e1dafd3d5	Stopped	t3.micro	-	us-east-1c	-
linux commands	i-004d001f1fb62f660	Stopped	t3.micro	-	us-east-1c	-
devopschallenge	i-0bb8dbdfef417e6a	Running	t3.micro	3/3 checks passed	us-east-1d	ec2-50-19-60-48.compute-1.amazonaws.com

```
bnarg@bnargava: MINGW64 ~/Downloads
$ ssh -i "devkey.pem" ec2-user@ec2-13-217-33-9.compute-1.amazonaws.com
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. see https://openssh.com/pq.html

#_
~\_##### Amazon Linux 2023
~~\#####
~~\#####
~~\#/_
~~V~'-'> https://aws.amazon.com/linux/amazon-linux-2023
~~~
~~~
~~~
~~~
Last login: Mon Aug 24 21:35:14 2026 from 18.206.107.28
[ec2-user@ip-172-31-37-66 ~]$ |
```

2. Install Docker.

```
ec2-user@ip-172-31-37-66:~$ sudo yum install docker
Running scriptlet: docker-25.0.16-1.amzn2023.0.4.x86_64
Installing : docker-25.0.16-1.amzn2023.0.4.x86_64
Running scriptlet: docker-25.0.16-1.amzn2023.0.4.x86_64
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch
Running scriptlet: docker-25.0.16-1.amzn2023.0.4.x86_64
Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch
Verifying : containerd-2.2.5-1.amzn2023.0.2.x86_64
Verifying : docker-25.0.16-1.amzn2023.0.4.x86_64
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
Verifying : libcgroupp-3.0-1.amzn2023.0.1.x86_64
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
Verifying : libnftnl-1.0.1-19.amzn2023.0.2.x86_64
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64
Verifying : pigz-2.5-1.amzn2023.0.4.x86_64
Verifying : runc-1.3.5-1.amzn2023.0.2.x86_64

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.4.x86_64
containerd-2.2.5-1.amzn2023.0.2.x86_64
iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
libcgroupp-3.0-1.amzn2023.0.1.x86_64
libnftnl-1.0.1-19.amzn2023.0.2.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64
runc-1.3.5-1.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-24 21:42:16 UTC; 28s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 28713 (dockerd)
     Tasks: 9
    Memory: 29.5M
       CPU: 341ms
    CGroup: /system.slice/docker.service
            └─28713 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Aug 24 21:42:15 ip-172-31-37-66.ec2.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Aug 24 21:42:15 ip-172-31-37-66.ec2.internal dockerd[28713]: time="2026-08-24T21:42:15.571929851Z" level=info msg="Starting up"
Aug 24 21:42:15 ip-172-31-37-66.ec2.internal dockerd[28713]: time="2026-08-24T21:42:15.639658195Z" level=info msg="Loading containers: start."
Aug 24 21:42:16 ip-172-31-37-66.ec2.internal dockerd[28713]: time="2026-08-24T21:42:16.092619516Z" level=info msg="Loading containers: done."
Aug 24 21:42:16 ip-172-31-37-66.ec2.internal dockerd[28713]: time="2026-08-24T21:42:16.126862893Z" level=info msg="Docker daemon" commit=6fdf0a6 containerd-snapshott
Aug 24 21:42:16 ip-172-31-37-66.ec2.internal dockerd[28713]: time="2026-08-24T21:42:16.127038601Z" level=info msg="Daemon has completed initialization"
Aug 24 21:42:16 ip-172-31-37-66.ec2.internal dockerd[28713]: time="2026-08-24T21:42:16.155230408Z" level=info msg="API listen on /run/docker.sock"
Aug 24 21:42:16 ip-172-31-37-66.ec2.internal systemd[1]: Started docker.service - Docker Application Container Engine.
[ec2-user@ip-172-31-37-66 ~]$ docker --version
Docker version 25.0.14, build 0babc007
[ec2-user@ip-172-31-37-66 ~]$
```

3. Instal Jenkins.

```
ec2-user@ip-172-31-37-66:~$ sudo yum install jenkins
Install 1 Package

Total download size: 96 M
Installed size: 96 M
Downloading Packages:
jenkins-2.568.2-1.noarch.rpm
29 MB/s | 96 MB 00:03

Total
29 MB/s | 96 MB 00:03

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing
  Running scriptlet: jenkins-2.568.2-1.noarch
  Installing : jenkins-2.568.2-1.noarch
  Running scriptlet: jenkins-2.568.2-1.noarch
  Verifying : jenkins-2.568.2-1.noarch

Installed:
jenkins-2.568.2-1.noarch

Complete!
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl start jenkins

[ec2-user@ip-172-31-37-66 ~]$
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-24 21:53:55 UTC; 46s ago
     Main PID: 29739 (java)
       Tasks: 44 (limit: 1059)
      Memory: 397.7M
         CPU: 21.054s
    CGroup: /system.slice/jenkins.service
            └─29739 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Aug 24 21:53:50 ip-172-31-37-66.ec2.internal jenkins[29739]: [LF]>
Aug 24 21:53:50 ip-172-31-37-66.ec2.internal jenkins[29739]: [LF]>
Aug 24 21:53:50 ip-172-31-37-66.ec2.internal jenkins[29739]: [LF]>
Aug 24 21:53:55 ip-172-31-37-66.ec2.internal jenkins[29739]: 2026-08-24 21:53:55.488+0000 [id=34] INFO Jenkins.InitReactorRunner$1.onAttained: Completed
Aug 24 21:53:55 ip-172-31-37-66.ec2.internal jenkins[29739]: 2026-08-24 21:53:55.498+0000 [id=50] INFO hudson.DownloadService$Downloadable#load: Obtained ts
Aug 24 21:53:55 ip-172-31-37-66.ec2.internal jenkins[29739]: 2026-08-24 21:53:55.499+0000 [id=50] INFO hudson.util.Retrier#start: Performed the action co
Aug 24 21:53:55 ip-172-31-37-66.ec2.internal jenkins[29739]: 2026-08-24 21:53:55.507+0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fu
Aug 24 21:53:55 ip-172-31-37-66.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Aug 24 21:54:00 ip-172-31-37-66.ec2.internal jenkins[29739]: 2026-08-24 21:54:00.596+0000 [id=70] WARNING h.n.DiskSpaceMonitorDescriptor#markNodeOffline
[ec2-user@ip-172-31-37-66 ~]$
```

4. install apache

```
ec2-user@ip-172-31-37-66:~$ sudo yum install httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
Running scriptlet: httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
Installing : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
Installing : httpd-core-2.4.68-1.amzn2023.0.1.x86_64
Installing : mod_http2-2.0.42-1.amzn2023.0.1.x86_64
Installing : mod_lua-2.4.68-1.amzn2023.0.1.x86_64
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Installing : httpd-2.4.68-1.amzn2023.0.1.x86_64
Running scriptlet: httpd-2.4.68-1.amzn2023.0.1.x86_64
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64
Verifying : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Verifying : httpd-2.4.68-1.amzn2023.0.1.x86_64
Verifying : httpd-core-2.4.68-1.amzn2023.0.1.x86_64
Verifying : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
Verifying : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch
Verifying : mod_http2-2.0.42-1.amzn2023.0.1.x86_64
Verifying : mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.3-1.amzn2023.0.2.x86_64          apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.68-1.amzn2023.0.1.x86_64
httpd-core-2.4.68-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch  httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch          mod_http2-2.0.42-1.amzn2023.0.1.x86_64      mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-24 21:57:07 UTC; 11s ago
     Docs: man:httpd.service(8)
   Main PID: 30395 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
   Tasks: 177 (limit: 1059)
  Memory: 14.2M
     CPU: 74ms
   CGroup: /system.slice/httpd.service
           └─30395 /usr/sbin/httpd -DFOREGROUND
           └─30396 /usr/sbin/httpd -DFOREGROUND
           └─30397 /usr/sbin/httpd -DFOREGROUND
           └─30434 /usr/sbin/httpd -DFOREGROUND
           └─30455 /usr/sbin/httpd -DFOREGROUND

Aug 24 21:57:07 ip-172-31-37-66.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 24 21:57:07 ip-172-31-37-66.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 24 21:57:07 ip-172-31-37-66.ec2.internal httpd[30395]: Server configured, listening on: port 80
[ec2-user@ip-172-31-37-66 ~]$
```

5.install nginx

```
ec2-user@ip-172-31-37-66:~$
[ec2-user@ip-172-31-37-66 ~]$ sudo nano /etc/nginx/nginx.conf
[ec2-user@ip-172-31-37-66 ~]$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl start nginx
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl status nginx
x nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: failed (Result: exit-code) since Mon 2026-08-24 22:02:03 UTC; 19s ago
     Process: 31413 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 31414 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 31415 ExecStart=/usr/sbin/nginx (code=exited, status=1/FAILURE)
    CPU: 60ms
Aug 24 22:02:01 ip-172-31-37-66.ec2.internal nginx[31415]: nginx: [emerg] bind() to 0.0.0.0:8080 failed (98: Address already in use)
Aug 24 22:02:01 ip-172-31-37-66.ec2.internal nginx[31415]: nginx: [emerg] bind() to [::]:8080 failed (98: Address already in use)
Aug 24 22:02:02 ip-172-31-37-66.ec2.internal nginx[31415]: nginx: [emerg] bind() to 0.0.0.0:8080 failed (98: Address already in use)
Aug 24 22:02:02 ip-172-31-37-66.ec2.internal nginx[31415]: nginx: [emerg] bind() to [::]:8080 failed (98: Address already in use)
Aug 24 22:02:02 ip-172-31-37-66.ec2.internal nginx[31415]: nginx: [emerg] bind() to 0.0.0.0:8080 failed (98: Address already in use)
Aug 24 22:02:02 ip-172-31-37-66.ec2.internal nginx[31415]: nginx: [emerg] bind() to [::]:8080 failed (98: Address already in use)
Aug 24 22:02:03 ip-172-31-37-66.ec2.internal nginx[31415]: nginx: [emerg] still could not bind()
Aug 24 22:02:03 ip-172-31-37-66.ec2.internal systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Aug 24 22:02:03 ip-172-31-37-66.ec2.internal systemd[1]: nginx.service: Failed with result 'exit-code'.
Aug 24 22:02:03 ip-172-31-37-66.ec2.internal systemd[1]: Failed to start nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-37-66 ~]$ sudo nano /etc/nginx/nginx.conf
[ec2-user@ip-172-31-37-66 ~]$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl start nginx
[ec2-user@ip-172-31-37-66 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-24 22:05:21 UTC; 27s ago
     Process: 31615 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 31616 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 31617 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 31618 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 59ms
    CGroup: /system.slice/nginx.service
            └─31618 "nginx: master process /usr/sbin/nginx"
              └─31619 "nginx: worker process"
                └─31620 "nginx: worker process"
Aug 24 22:05:21 ip-172-31-37-66.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Aug 24 22:05:21 ip-172-31-37-66.ec2.internal nginx[31616]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 24 22:05:21 ip-172-31-37-66.ec2.internal nginx[31616]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 24 22:05:21 ip-172-31-37-66.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-37-66 ~]$
```

WhatsApp

kanchannudra5000-a11y/

cloud challenges

Launch an instance | EC2

Instances | EC2 | us-east-1

Welcome to nginx!

13.217.333.81

Agoda.com Agoda.com Maps TypeTest.io Bhargava Karli CCB... Frontend Projects... Sample Resume Te... Web 3 Concept... Interviews Guidelin...

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

6. Install Apache Tomcat.

```

ec2-user@ip-172-31-37-66/tmp
-rwxr-x--- 1 root root 1922 Aug 13 16:59 configtest.sh
-rwxr-x--- 1 root root 8319 Aug 13 16:59 daemon.sh
-rw-r----- 1 root root 2091 Aug 13 16:59 digest.bat
-rwxr-x--- 1 root root 1965 Aug 13 16:59 digest.sh
-rw-r----- 1 root root 3606 Aug 13 16:59 makebase.bat
-rwxr-x--- 1 root root 3382 Aug 13 16:59 makebase.sh
-rw-r----- 1 root root 2096 Aug 13 16:59 migrate.bat
-rwxr-x--- 1 root root 1970 Aug 13 16:59 migrate.sh
-rw-r----- 1 root root 3382 Aug 13 16:59 setclasspath.bat
-rwxr-x--- 1 root root 3941 Aug 13 16:59 setclasspath.sh
-rw-r----- 1 root root 2020 Aug 13 16:59 shutdown.bat
-rwxr-x--- 1 root root 1902 Aug 13 16:59 shutdown.sh
-rw-r----- 1 root root 2022 Aug 13 16:59 startup.bat
-rwxr-x--- 1 root root 1904 Aug 13 16:59 startup.sh
-rw-r----- 1 root root 52518 Aug 13 16:59 tomcat-juli.jar
-rw-r----- 1 root root 476797 Aug 13 16:59 tomcat-native.tar.gz
-rw-r----- 1 root root 3623 Aug 13 16:59 tool-wrapper.bat
-rwxr-x--- 1 root root 4600 Aug 13 16:59 tool-wrapper.sh
-rw-r----- 1 root root 2026 Aug 13 16:59 version.bat
-rwxr-x--- 1 root root 1908 Aug 13 16:59 version.sh
[ec2-user@ip-172-31-37-66 tmp]$ sudo /opt/tomcat/bin/startup.sh
sudo: /opt/tomcat/bin/startup.sh: command not found
[ec2-user@ip-172-31-37-66 tmp]$ sudo /opt/tomcat/bin/startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[ec2-user@ip-172-31-37-66 tmp]$ ps -ef | grep tomcat
root      32557      1  21 22:21 ?        00:00:04 /usr/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manager=org.a
pache.juli.ClassLoaderLogManager -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.webresources -Dsun.io.useCanonCaches=false -Dorg.ap
ache.catalina.security.SecurityListener.Umask=0027 --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.bas
e/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNN
AMED -classpath /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar -Dcatalina.base=/opt/tomcat -Dcatalina.home=/opt/tomcat -Djava.io.tmpdir=/opt/tomcat/temp
org.apache.catalina.startup.Bootstrap start
ec2-user  32577  31923  0 22:21 pts/1    00:00:00 grep --color=auto tomcat
[ec2-user@ip-172-31-37-66 tmp]$ curl http://localhost:8080
<html><head><meta http-equiv='refresh' content='1;url=/login?from=%2F'><script id='redirect' data-redirect-url='/login?from=%2F' src='/static/5dcf6fb0/scripts/redirec
t.js'></script></head><body style='background-color:white; color:white;'>
Authentication required
<!--
You are authenticated as: anonymous
Groups that you are in:
    anonymous
Permission you need to have (but didn't): hudson.model.Hudson.Administer
-->
</body></html>
[ec2-user@ip-172-31-37-66 tmp]$
[ec2-user@ip-172-31-37-66 tmp]$

```